

Input File Description

Program: pp.x / PWscf / Quantum ESPRESSO (version: 7.5)

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INTRODUCTION

Purpose of pp.x: data analysis and plotting.

The code performs two steps:

- (1) reads the output produced by pw.x, extracts and calculates the desired quantity/quantities (rho, V, ...)
- (2) writes the desired quantity to file in a suitable format for various types of plotting and various plotting programs

The input data of this program is read from standard input or from file and has the following format:

NAMELIST **&INPUTPP**
containing the variables for step (1), followed by

NAMELIST **&PLOT**
containing the variables for step (2)

The two steps can be performed independently. In order to perform only step (2), leave namelist **&INPUTPP** blank. In order to perform only step (1), do not specify namelist **&PLOT**

Intermediate results from step 1 can be saved to disk (see variable [filplot](#) in **&INPUTPP**) and later read in step 2. Since the file with intermediate results is formatted, it can be safely transferred to a different machine. This also allows plotting of a linear combination (for instance, charge differences) by saving two intermediate files and combining them (see variables [weight](#) and [filepp](#) in **&PLOT**)

All output quantities are in ATOMIC (RYDBERG) UNITS unless otherwise explicitly specified.
All charge densities integrate to the NUMBER of electrons not to the total charge.
All potentials have the dimension of an energy (e*V, not V).

Namelist: &INPUTPP

title	CHARACTER
<i>Default:</i>	''
reprinted on output, CUB files and other places	
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prefix	CHARACTER
prefix of files saved by program pw.x	
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outdir	CHARACTER
<i>Default:</i>	value of the ESPRESSO_TMPDIR environment variable if set; current directory ('./') otherwise
directory containing the input data, i.e. the same as in pw.x	

filplot CHARACTER*Default:* 'tmp.pp'

file "filplot" contains the quantity selected by plot_num.
Typically this is a temporary file and may be ignored,
unless you want to save it for further processing

plot_num INTEGER*Default:* -1

Selects what to save in filplot:

- 1 = do not do anything, skip step (1) (see header)
- 0 = electron (pseudo-)charge density
- 1 = total potential $V_{\text{bare}} + V_{\text{H}} + V_{\text{xc}}$
- 2 = local ionic potential V_{bare}
- 3 = local density of states at specific energy or grid of energies
(number of states per volume, in bohr³, per energy unit, in Ry)
- 4 = local density of electronic entropy
- 5 = STM images
Tersoff and Hamann, [PRB 31, 805 \(1985\)](#)
- 6 = spin polarization ($\rho(\text{up}) - \rho(\text{down})$)
- 7 = contribution of selected wavefunction(s) to the
(pseudo-)charge density. For norm-conserving PPs,
 $|\psi|^2$ (ψ =selected wavefunction). Noncollinear case:
contribution of the given state to the charge or
to the magnetization along the direction indicated
by spin_component (0 = charge, 1 = x, 2 = y, 3 = z)
- 8 = electron localization function (ELF)
- 9 = charge density minus superposition of atomic densities
- 10 = integrated local density of states (ILDOS)
from [emin](#) to [emax](#) (emin, emax in eV)
if [emax](#) is not specified, [emax](#)= E_{fermi}
- 11 = the $V_{\text{bare}} + V_{\text{H}}$ potential
- 12 = the sawtooth electric field potential (if present)
- 13 = the noncollinear magnetization.
- 17 = all-electron valence charge density
can be performed for PAW calculations only
requires a very dense real-space grid!
- 18 = The exchange and correlation magnetic field in the noncollinear case
- 19 = Reduced density gradient
(J. Chem. Theory Comput. 7, 625 (2011), [doi:10.1021/ct100641a](#))
Set the isosurface between 0.3 and 0.6 to plot the
non-covalent interactions (see also plot_num = 20)
- 20 = Product of the electron density (charge) and the second
eigenvalue of the electron-density Hessian matrix;
used to colorize the RDG plot (plot_num = 19)
- 21 = all-electron charge density (valence+core).
For PAW calculations only; requires a very dense real-space grid.
- 22 = kinetic energy density
- 23 = the charge density of states between emin & emax
- 123 = DORI: density overlap regions indicator
([doi: 10.1021/ct500490b](#)) Implemented by D. Yang & Q.Liu
- 24 = Reconstructed all-electron charge
- 25 = Squared modulus of the Hubbard projector function of DFT+U

IF $plot_num = 0$ or 9 :

Options for total charge ($plot_num=0$) or for total minus atomic charge ($plot_num=9$):

spin_component INTEGER
 Default: 0
 0 = total charge (default value),
 1 = spin up charge,
 2 = spin down charge.

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ELSEIF $plot_num=1$:

Options for total potential ($plot_num=1$):

spin_component INTEGER
 Default: 0
 0 = spin averaged potential (default value),
 1 = spin up potential,
 2 = spin down potential.

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ELSEIF $plot_num=3$:

Options for LDOS ($plot_num=3$): LDOS is plotted on grid [e_{min} , e_{max}] with spacing δ_e .

e_{min} REAL
 Default: e_{fermi}
 lower boundary of energy grid (in eV).
 Defaults to Fermi energy.

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e_{max} REAL
 Status: OPTIONAL
 upper boundary of energy grid (in eV).
 Defaults to Fermi energy.

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δ_e REAL
 Default: 0.1
 Status: OPTIONAL
 spacing of energy grid (in eV).

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degauss_ldos REAL
 Default: degauss (converted to eV)
 Status: OPTIONAL
 broadening of energy levels for LDOS (in eV).
 Defaults to broadening degauss specified for electronic smearing
 in pw.x calculation.

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use_gauss_ldos LOGICAL
 Default: .false.
 Status: OPTIONAL
 If .true., gaussian broadening ($ngauss=0$) is used for LDOS calculation.
 Defaults .false., in which case the broadening scheme

of the pw.x calculation will be used.

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ELSEIF *plot_num=5* :

Options for STM images (plot_num=5):

sample_bias REAL
the bias of the sample (Ry) in stm images

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ELSEIF *plot_num=7* :

Options for $|\psi|^2$ (plot_num=7):

kpoint(i), i=1,2 INTEGER
Unpolarized and noncollinear case:
 k-point(s) to be plotted
LSDA:
 k-point(s) and spin polarization to be plotted
 (spin-up and spin-down correspond to different k-points!)

To plot a single kpoint ikpt, specify kpoint=ikpt or kpoint(1)=ikpt
To plot a range of kpoints [imin, imax], specify kpoint(1)=imin and kpoint(2)=imax

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kband(i), i=1,2 INTEGER
Band(s) to be plotted.

To plot a single band ibnd, specify kband=ibnd or kband(1)=ibnd
To plot a range of bands [imin, imax], specify kband(1)=imin and kband(2)=imax

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lsign LOGICAL
if true and k point is Gamma, plot $|\psi|^2$ sign(ψ)

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spin_component(i), i=1,2 INTEGER
 Default: 0
 Status: OPTIONAL

Noncollinear case only:
plot the contribution of the given state(s) to the charge
or to the magnetization along the direction(s) indicated
by spin_component:
 0 = charge (default),
 1 = x,
 2 = y,
 3 = z.

Ignored in unpolarized or LSDA case

To plot a single component ispin, specify spin_component=ispin or spin_component(1)=ispin
To plot a range of components [imin, imax], specify spin_component(1)=imin and spin_component(2)=imax

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ELSEIF *plot_num=10* :

Options for ILDOS (plot_num=10):

emin REAL
lower energy boundary (in eV)

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emax REAL

upper energy boundary (in eV),
i.e. compute ILDOS from [emin](#) to [emax](#)

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spin_component INTEGER

Default: 0

for LSDA case only: plot the contribution to ILDOS of
0 = spin-up + spin-down (default)
1 = spin-up only
2 = spin-down only

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ELSEIF *plot_num*=13 :

Options for noncollinear magnetization (plot_num=13):

spin_component INTEGER

Default: 0

0 = absolute value (default value)
1 = x component of the magnetization
2 = y component of the magnetization
3 = z component of the magnetization

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ELSEIF *plot_num*=17 :

Options for reconstructed charge density (plot_num=17):

spin_component INTEGER

Default: 0

0 = total charge (default value),
1 = spin up charge,
2 = spin down charge.

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ELSEIF *plot_num*=22 :

Options for kinetic energy density (plot_num=22), LSDA case only:

spin_component INTEGER

Default: 0

0 = total density (default value),
1 = spin up density,
2 = spin down density.

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ELSEIF *plot_num*=25 :

Options for kinetic energy density (plot_num=22), LSDA case only:

nc INTEGER

Default: 1

nc(1), nc(2), and nc(3) indicate the size of the
supercell for the visualization of the squared modulus of
the Hubbard projector functions.
Note, this supercell must match the k-points grid size
otherwise the resulting WFs are wrong.

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n0 INTEGER

Default: 0

n0(1), n0(2), and n0(3) indicate the shift
of the coordinate frame in units of the

lattice vectors for a better visualization
of the Hubbard projector functions

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Namelist: &PLOT

nfile INTEGER
 Default: 1
 Status: OPTIONAL
 the number of data files to read

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filepp(i), i=1,nfile CHARACTER
 Default: filepp(1)='tmp.pp'
 nfile = 1 : file containing the quantity to be plotted
 (should be the same as [filplot](#))
 nfile > 1 : see [weight](#)

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weight(i), i=1,nfile REAL
 Default: weight(1)=1.0
 weighing factors: assuming that rho(i) is the quantity
 read from filepp(i), the quantity that will be plotted is:
 weight(1)*rho(1) + weight(2)*rho(2) + weight(3)*rho(3) + ...

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BEWARE: atomic coordinates are read from the first file;
 if their number is different for different files,
 the first file must have the largest number of atoms

iflag INTEGER
 0 = 1D plot of the spherical average
 1 = 1D plot
 2 = 2D plot
 3 = 3D plot
 4 = 2D polar plot on a sphere

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output_format INTEGER
 (ignored on 1D plot)
 0 = format suitable for gnuplot (1D)
 1 = obsolete format no longer supported
 2 = format suitable for plotrho (2D)
 3 = format suitable for XCRYSDEN (2D or user-supplied 3D region)
 4 = obsolete format no longer supported
 5 = format suitable for XCRYSDEN (3D, using entire FFT grid)
 6 = format as gaussian cube file (3D)
 (can be read by many programs)
 7 = format suitable for gnuplot (2D) x, y, f(x,y)

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fileout CHARACTER
 Default: standard output

name of the file to which the plot is written

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interpolation CHARACTER

Default: 'fourier'

Type of interpolation:

'fourier'

'bspline' :
(EXPERIMENTAL)

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IF *iflag* = 0 or 1 :

the following variables are REQUIRED:

e1(i), i=1,3 REAL

3D vector which determines the plotting line (in alat units)

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x0(i), i=1,3 REAL

3D vector, origin of the line (in alat units)

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nx INTEGER

number of points in the line:

$\rho(i) = \rho(x_0 + e_1 * (i-1)/(nx-1)), i=1, nx$

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ELSEIF *iflag* = 2 :

the following variables are REQUIRED:

e1(i), e2(i), i=1,3 REAL

3D vectors which determine the plotting plane (in alat units)

BEWARE: **e1** and **e2** must be orthogonal

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x0(i), i=1,3 REAL

3D vector, origin of the plane (in alat units)

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nx, ny INTEGER

Number of points in the plane:

$\rho(i,j) = \rho(x_0 + e_1 * (i-1)/(nx-1) + e_2 * (j-1)/(ny-1)), i=1,nx ; j=1,ny$

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ELSEIF *iflag* = 3 :

the following variables are OPTIONAL:

e1(i), e2(i), e3(i), i=1,3 REAL

3D vectors which determine the plotting parallelepiped
(if present, must be orthogonal)

[e1](#), [e2](#), and [e3](#) are in alat units !

[\[Back to Top\]](#)**x0(i), i=1,3** REAL

3D vector, origin of the parallelepiped

[x0](#) is in alat units ![\[Back to Top\]](#)**nx, ny, nz** INTEGER

Number of points in the parallelepiped:

$$\text{rho}(i,j,k) = \text{rho}(x_0 + e_1 * (i-1)/nx + e_2 * (j-1)/ny + e_3 * (k-1)/nz),$$

$$i = 1, nx ; j = 1, ny ; k = 1, nz$$

- If [output_format](#) = 3 (XCRYSDEN), the above variables are used to determine the grid to plot.
- If [output_format](#) = 5 (XCRYSDEN), the above variables are ignored, the entire FFT grid is written in the XCRYSDEN format - works for any crystal axis (VERY FAST)
- If [e1](#), [e2](#), [e3](#), [x0](#) are present, and [e1](#), [e2](#), [e3](#) are parallel to xyz and parallel to crystal axis, a subset of the FFT grid that approximately covers the parallelepiped defined by [e1](#), [e2](#), [e3](#), [x0](#), is written - untested, might be obsolete
- Otherwise, the required 3D grid is generated from the Fourier components (may be VERY slow)

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Radius of the sphere (alat units), centered at (0,0,0)

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Number of points in the polar plane:

$$\text{phi}(i) = 2 \pi * (i - 1)/(nx-1), i=1, nx$$

$$\text{theta}(j) = \pi * (j - 1)/(ny-1), j=1, ny$$
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